

GORYACHINSK, Russian Federation

WMO: 307310

Lat: 52.986N

Lon: 108.291E

Elev: 491

StdP: 95.56

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.1	-26.8	-32.2	0.2	-28.8	-29.8	0.3	-26.6	8.1	-12.9	7.2	-13.2	0.9	130	0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	24.1	17.3	22.1	16.6	20.5	15.9	18.2	22.2	17.3	20.9	16.5	19.8	1.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	12.6	20.3	15.8	11.9	19.3	15.0	11.3	18.5	53.5	22.4	50.7	21.0	48.2	19.9	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.7	6.0	5.3	-31.0	28.6	3.5	2.6	-33.5	30.4	-35.6	32.0	-37.5	33.4	-40.1	35.3		
			-31.1	20.0	3.5	0.9	-33.6	20.6	-35.7	21.1	-37.6	21.6	-40.2	22.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.1	-16.8	-15.9	-8.9	-0.1	5.8	11.7	15.9	15.3	9.0	1.8	-6.0	-11.3
	DBStd	12.06	6.02	5.72	5.95	3.90	3.29	3.02	2.65	2.64	3.31	3.58	5.35	5.04
	HDD10.0	4057	832	727	586	304	138	15	0	1	57	255	481	662
	HDD18.3	6667	1090	960	844	554	390	199	86	101	279	513	731	920
	CDD10.0	446	0	0	0	0	7	67	182	163	28	0	0	0
	CDD18.3	16	0	0	0	0	0	1	9	6	0	0	0	0
	CDH23.3	93	0	0	0	0	1	13	45	34	1	0	0	0
	CDH26.7	9	0	0	0	0	0	1	4	4	0	0	0	0
Wind	WSAvg	2.1	2.1	1.7	1.8	1.9	2.0	1.7	1.7	1.8	2.3	2.4	3.1	3.1
Precipitation	PrecAvg	399	14	5	5	15	26	43	69	72	43	20	41	41
	PrecMax	533	57	10	26	33	48	113	181	136	99	62	80	113
	PrecMin	272	5	1	0	1	5	5	16	18	9	4	8	8
	PrecStd	75	12	2	5	8	11	24	44	33	23	12	19	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.1	0.1	6.9	13.2	19.5	25.4	27.0	26.7	21.6	13.2	7.2	0.7
		MCWB	-2.5	-2.8	2.2	6.8	11.1	15.3	17.9	17.5	13.9	7.7	3.1	-1.9
	2%	DB	-3.5	-2.5	4.0	10.0	16.1	21.4	24.4	23.6	18.3	10.5	5.1	-1.5
		MCWB	-5.1	-4.5	0.5	4.7	9.1	14.7	17.8	17.2	13.5	6.1	2.0	-3.7
	5%	DB	-6.3	-5.1	1.7	7.5	13.9	18.9	22.5	21.6	16.3	8.8	3.2	-3.0
		MCWB	-7.9	-6.6	-1.1	3.4	8.0	13.8	17.2	16.6	11.9	5.4	0.6	-4.9
	10%	DB	-8.5	-7.4	-0.4	5.7	12.1	17.1	20.8	20.2	14.4	7.3	1.5	-4.6
		MCWB	-9.7	-8.6	-2.5	2.2	7.1	12.6	16.5	15.8	10.9	4.2	-0.9	-6.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.5	-2.6	2.5	7.1	11.7	17.3	20.0	19.1	15.8	8.6	3.6	-1.5
		MCDB	-1.2	-0.2	6.3	12.9	17.8	21.4	24.9	23.9	19.5	12.6	6.7	0.5
	2%	WB	-5.2	-4.6	0.7	5.0	9.9	15.6	18.6	17.9	13.9	6.7	2.2	-3.6
		MCDB	-3.6	-2.6	3.7	9.3	15.1	19.8	22.9	22.0	17.5	9.6	4.9	-1.7
	5%	WB	-7.9	-6.6	-0.9	3.7	8.7	14.3	17.7	17.1	12.6	5.7	0.7	-4.8
		MCDB	-6.5	-5.1	1.5	7.2	13.2	18.1	21.3	20.8	15.6	8.6	3.1	-3.2
	10%	WB	-9.7	-8.6	-2.5	2.5	7.5	13.2	16.9	16.3	11.2	4.4	-0.8	-6.1
		MCDB	-8.6	-7.5	-0.4	5.5	11.5	16.6	20.3	19.6	14.0	7.1	1.4	-4.5
Mean Daily Temperature Range	5% DB	MDBR	6.9	8.3	8.8	8.3	9.2	8.8	7.7	7.6	7.6	6.9	6.0	5.5
		MCDBR	6.9	9.2	9.5	11.1	12.6	12.3	10.6	10.8	10.5	9.6	7.1	5.7
		MCWBR	6.0	7.6	6.8	7.1	7.4	6.7	5.4	5.8	6.1	6.9	5.7	5.0
	5% WB	MCDBR	6.8	9.2	9.3	10.6	11.2	10.4	8.7	9.3	9.0	9.3	7.0	5.6
		MCWBR	6.0	7.7	6.9	7.1	7.1	6.3	4.9	5.3	5.6	6.9	5.8	5.0
Clear-Sky Solar Irradiance	taub	0.262	0.290	0.318	0.365	0.429	0.440	0.428	0.385	0.337	0.308	0.278	0.259	
	taud	2.094	2.079	2.121	2.241	2.121	2.209	2.289	2.393	2.474	2.422	2.158	2.052	
	Ebn at Noon	703	799	858	864	820	815	818	838	844	792	676	617	
	Edh at Noon	71	100	120	122	145	135	123	105	85	72	66	62	
All-Sky Solar Radiation	RadAvg	1.10	2.12	3.58	4.88	5.70	6.04	5.54	4.67	3.43	2.05	1.07	0.75	
	RadStd	0.08	0.11	0.13	0.31	0.35	0.38	0.44	0.22	0.29	0.19	0.09	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (12 neighbors)	N/A	N/A	-2.08	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page